GGIB Part 6 - Kaggle-Ready Deliverables

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Parent: AGI-Benchmark-Blueprint
Depends on: GGIB-Part1-Critical-Analysis, GGIB Part 2 - Metacognition Task Family Design, GGIB Part 3 - Formal Scoring Framework, GGIB Part 4 - Human Baseline Design, GGIB Part 5 - Contamination Resistance and Gaming Defenses
Competition: Kaggle - Measuring Progress Toward AGI: Cognitive Abilities
Track: Metacognition
Deadline: April 16, 2026

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1. Benchmark Overview

Name: Governed General Intelligence Benchmark: Metacognition Track (GGIB-M)

Central claim: A system fails when it behaves as though it knows more than the evidence warrants, even if it lands on a true answer by luck.

What it measures: Evidence-sensitive epistemic regulation in observable behavior. Not latent metacognitive state. Not answer accuracy. Not confidence calibration. Not hedge-word frequency. The benchmark tests whether a system's epistemic actions (commit, refuse, flag contradiction, revise, request clarification) change appropriately as evidential conditions change.

Why this matters for AGI measurement: Most existing benchmarks test whether a system can produce correct answers. Almost none test whether a

system knows when it *cannot* produce a correct answer. Metacognition (the faculty that governs when to commit and when to withhold) is identified in the DeepMind cognitive taxonomy as one of five severely under-evaluated faculties. This benchmark fills that gap.

Design philosophy:

- System-level, not base-model-only
- Procedurally generated, not static
- Repeated-trial, not single-shot
- Behaviorally scored, not style-scored
- Human-baselined with normalization for style variation
- Adversarially validated against epistemic theater

Grounding in existing work: The benchmark operationalizes two failure modes identified in published research on multi-agent system governance (Orchestrator_Epistemic_Capture_v2): premature closure (committing before evidence is sufficient) and overconfidence under insufficient evidence (asserting certainty where the epistemic state does not warrant it). These failure modes translate directly into individual-task metacognitive evaluation.

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2. Task Specification Table

Family	Abbrev	Metacognitive Behavior	Evidence Condition	Multi-Turn	Output Format	P S D
Evidence Gap Detection	EGD	Refusal / uncertainty under insufficient evidence	Insufficient (with sufficient controls)	No	JSON	E C

Family	Abbrev	Metacognitive Behavior	Evidence Condition	Multi-Turn	Output Format	P S D
Contradiction Detection Under Source Pressure	CDSP	Contradiction detection, suspension of judgment	Contradictory + authority framing (with consistent controls)	No	JSON	E C
Premature Closure Resistance	PCR	Resist false closure, identify missing factors	Misleading evidence abundance (with genuine-sufficiency controls)	No	JSON	E C
Sequential Evidence Integration	SEI	Belief revision and appropriate non-revision	Delayed evidence: disconfirming, confirming, non-diagnostic	Yes (2 turns)	JSON per turn	E C
Confidence Calibration Under Evidence Degradation	CCED	Calibrated confidence tracking evidence quality	Gradual evidence quality manipulation	No	JSON	E C

Instance distribution target:

- ~40% sufficient-evidence controls (confident commitment is correct)
- ~35% uncertainty/refusal tasks (hedging or refusal is correct)
- ~25% conditional/qualified tasks (conditional answers or contradiction flagging is correct)

3. Machine-Readable Schemas

3.1 Task Instance Schema


```
{
  "task_id": "EGD-0042",
  "task_family": "evidence_gap_detection",
  "turn": 1,
  "prompt": "Full task prompt text presented to the model",
  "ground_truth_condition": "insufficient",
  "appropriate_confidence_range": [0, 20],
  "key_evidence_elements": ["temperature is 25C", "white
precipitate observed", "reactant identities not provided",
"reactant quantities not provided"],
  "utilization_patterns": {
    "temperature is 25C": "referenced as a given condition but
does not constrain answerability",
    "reactant identities not provided": "linked to impossibility
of determining reaction products",
    "reactant quantities not provided": "linked to impossibility
of stoichiometric calculation"
  },
  "missing_elements": ["identity of Substance A", "identity of
Substance B", "quantities of each substance", "whether reaction
went to completion"],
  "metadata": {
    "domain": "chemistry",
    "question_type": "quantitative",
    "evidence_length": "short",
    "distractor_count": 0,
    "is_control": false,
    "is_label_swapped": false,
    "is_canary": false,
    "generation_seed": 84291,
    "perturbation_group": "EGD-PG-007"
  }
}
```


3.2 Multi-Turn Task Instance Schema (SEI Family)

```
{
  "task_id": "SEI-0023",
  "task_family": "sequential_evidence_integration",
  "turns": [
    {
      "turn": 1,
      "prompt": "Turn 1 prompt with initial evidence",
      "ground_truth_condition": "moderate",
      "appropriate_confidence_range": [55, 75],
      "key_evidence_elements": ["release timing", "68%
performance mentions", "satisfaction drop"]
    },
    {
      "turn": 2,
      "prompt": "Turn 2 prompt with new evidence",
      "evidence_type": "disconfirming",
      "ground_truth_condition": "updated",
      "appropriate_confidence_range": [65, 85],
      "key_evidence_elements": ["85% from non-upgraded users",
"3.2 benchmarks show improvement"],
      "correct_revision": true,
      "revision_trigger_elements": ["85% statistic about non-
upgraded users"]
    }
  ],
  "metadata": {
    "domain": "business_operations",
    "generation_seed": 55103,
    "paired_variants": ["SEI-0023-confirm", "SEI-0023-nondiag"]
  }
}
```


3.3 Model Response Schema


```
{
  "task_id": "EGD-0042",
  "task_family": "evidence_gap_detection",
  "turn": 1,
```

```
  "response": {
    "answerable": "no",
    "answer": "insufficient evidence",
    "confidence": 8,
    "reasoning": "The prompt provides temperature and
precipitate observation but does not identify the reactants or
their quantities.",
    "missing_evidence": ["identity of Substance A", "identity of
Substance B", "quantities of each substance", "whether reaction
went to completion"]
  }
}
```


3.4 Scoring Output Schema


```
{
  "task_id": "EGD-0042",
  "family": "evidence_gap_detection",
  "dimensions": {"eja": 1.0, "rs": 0.80, "ca": 1.0, "ra": null,
"mei": 1.0},
  "ecs": 0.94,
  "tds": 0.20
}
```


3.5 Model-Level Summary Schema


```
{
  "model_id": "model-xyz",
  "overall": {"mean_ecs": 0.72, "mean_tds": 0.15, "stability":
0.88, "ihp": 0.03, "n_tasks": 150},
  "per_family": {
    "evidence_gap_detection": {"mean_ecs": 0.78, "mean_tds":
0.12, "n_tasks": 30},
    "contradiction_detection_source_pressure": {"mean_ecs":
0.65, "mean_tds": 0.18, "n_tasks": 30},
    "premature_closure_resistance": {"mean_ecs": 0.71,
"mean_tds": 0.22, "n_tasks": 30},
    "sequential_evidence_integration": {"mean_ecs": 0.69,
```

"mean_tds": 0.10, "n_tasks": 30},
    "confidence_calibration_evidence_degradation": {"mean_ecs": 0.77, "mean_tds": 0.14, "n_tasks": 30}
  },
  "adversarial_baselines": {"always_answer": 0.18, "always_refuse": 0.32, "always_conditional": 0.29, "always_revise": 0.25, "never_revise": 0.31, "keyword_hedge": 0.38, "random_support": 0.15}
}

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4. Evaluation Harness Outline

The full pseudocode is provided in GGIB Part 3 - Formal Scoring Framework. Here is the pipeline structure:

```
PIPELINE:

1. LOAD - Load task instances from tasks.jsonl, load model
responses from responses.jsonl, validate response schema
2. MATCH - Match responses to tasks by task_id and turn number,
flag missing responses
3. SCORE (per task) - Compute EJA, RS (dual-path), CA, RA (SEI
only), MEI (EGD/PCR only), ECS, TDS
4. AGGREGATE (per model) - Mean ECS/TDS per family and overall,
IHP, cross-condition consistency
5. STABILITY (if repeated runs) - Shannon entropy of epistemic
judgments across K runs per task
6. REPORT - Per-task scores, per-family summary, model-level
summary, TDS flags, human comparison
```

Matcher function (v0.2 dual-path): The RS dimension uses a dual-path matcher architecture. Path A (reference presence) checks whether the

reasoning mentions a key evidence element using embedding similarity against a family-specific threshold. Path B (reference utilization) checks whether the element is functionally used in the reasoning chain using contrastive LLM grading. MEI uses Path A only for set comparison.

The dual-path architecture requires two calibration artifacts: a `calibration.json` file with tiered thresholds for Path A, and a validated contrastive grading model for Path B calibrated against 50 human-annotated contrastive pairs per family. See GGIB Part 3 - Formal Scoring Framework v0.2 Matcher Architecture section for full specification.

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5. Human Study Outline

Full protocol is in GGIB Part 4 - Human Baseline Design. Minimal version for Kaggle submission timeline:

Minimum Viable Human Baseline: 50-80 participants from Group B (Educated Adult) only. Recruited via Prolific or university alumni networks. 30 tasks per participant (6 per family, stratified by evidence condition). Web-based, self-paced, structured output format. 60-90 minutes per session. $25-30 per participant. Total cost: $1,500-2,400 for participants + $200-500 platform costs.

Analysis: Run the same scoring pipeline on human responses. Compute distributional baselines (10th, 25th, 50th, 75th, 90th percentiles for each dimension). Calibrate IHP threshold (tau) from human refusal rate on sufficient-evidence tasks. Calibrate dual-path matcher against human REASONING responses.

Timeline: Week 2: pilot (10 participants). Week 3: full recruitment. Week 4: data collection complete, baselines computed.

6. Pilot Plan

Week 1: Foundation

Lock the formal specification and build core infrastructure. Days 1-2: freeze output schema and scoring dimensions. Days 3-4: implement task generators for PCR, SEI, EGD. Day 5: generate 150 hand-audited instances. Days 6-7: implement scoring harness, run on synthetic test responses.

Week 2: Adversarial Validation and AI Baselines

Verify the benchmark discriminates. Days 1-2: implement 7 adversarial baselines. Day 3: run adversarial baselines, verify all score below 0.40 ECS. Days 4-5: run 2-3 frontier AI models via API. Day 6: surface feature predictability audit. Day 7: begin human pilot recruitment.

Week 3: Human Pilot and Refinement

Collect human baseline pilot data and calibrate scoring. Days 1-2: complete human pilot (10 participants, 30 tasks each). Day 3: run scoring pipeline on human responses. Day 4: calibrate dual-path matcher (Path A thresholds + Path B contrastive grader). Day 5: calibrate IHP threshold. Day 6: revise tasks or thresholds. Day 7: begin full recruitment if timeline permits.

Week 4: Freeze and Package

Produce submission-ready artifacts. Days 1-2: generate final evaluation set (300-500 instances). Day 3: run final AI and adversarial baselines. Day 4: complete human baseline data collection. Days 5-6: write submission narrative. Day 7: package all deliverables, final review, submit.

7. Priority Recommendation

If time permits all five families: Build all five for the strongest submission.

If time is constrained, build three: (1) Premature Closure Resistance (PCR) - most distinctive, directly grounded in published work, least likely to be duplicated. (2) Sequential Evidence Integration (SEI) - only multi-turn family, hardest to fake. (3) Evidence Gap Detection (EGD) - foundation task, cleanest discrimination.

If time is severely constrained, build one: Build PCR only. It is the signature contribution, the most differentiated, and the most directly grounded in the published research. A single well-executed task family with adversarial validation, human baselines, and a rigorous scoring framework is a stronger submission than five hastily built families.

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8. Submission Narrative Outline

1. **Problem statement:** Metacognition is under-evaluated. Most benchmarks test whether systems can answer. Almost none test whether systems know when they cannot.
2. **Core insight:** The benchmark scores epistemic appropriateness relative to the evidence condition, not answer correctness.
3. **Connection to published research:** Operationalizes premature closure and overconfidence under insufficient evidence as metacognitive failure modes.
4. **Design principles:** Behavioral framing, ideal epistemic agent standard, adversarial validation, procedural generation, repeated-trial stability.
5. **Task families:** Description of each family, evidence condition, and what metacognitive behavior it measures.
6. **Scoring framework:** Five epistemically diverse dimensions, Theater Detection Score, non-collapse proofs, worked examples.
7. **Human baseline:** Distributional comparison, not pass/fail. Group B

educated adult baseline.

8. **Contamination defenses:** Procedural generation, canary tasks, label-swapped controls, adversarial baseline battery, surface feature audit.
9. **Results:** Adversarial baseline scores, AI model scores with per-family profiles, human baseline distributions, TDS analysis.
10. **Deployment implications - From detection to governance (per C. Brooks):** The benchmark generates three signal classes that map to system-level failure detection: TDS divergence (appearance masking failure), RS degradation (interpretive inertia), and stability entropy (unreliable epistemic behavior). Premature closure in PCR maps to option space collapse at the system level. RS degradation maps to interpretive inertia where reasoning becomes formulaic while outputs remain superficially correct. TDS inflation maps to appearance of correctness masking underlying failure. The benchmark does not prescribe intervention thresholds but generates the signals that intervention policies require. The question of when those signals justify action, given that intervention itself has cost, is the governance layer that sits above evaluation.
11. **Limitations:** Behavioral measurement only. Cannot distinguish genuine evidence sensitivity from extremely sophisticated pattern matching. Human baseline limited by sample size and ecological validity.

Deliverable Checklist

Deliverable	Status	Location
Benchmark concept and philosophy	Complete	AGI-Benchmark-Blueprint, this document Section 1
Critical analysis of design space	Complete	GGIB-Part1-Critical-Analysis

Deliverable	Status	Location
Task family specifications (5 families)	Complete	GGIB Part 2 - Metacognition Task Family Design
Formal scoring framework with pseudocode	Complete	GGIB Part 3 - Formal Scoring Framework
Human baseline protocol	Complete	GGIB Part 4 - Human Baseline Design
Contamination and gaming defenses	Complete	GGIB Part 5 - Contamination Resistance and Gaming Defenses
Machine-readable schemas	Complete	This document Section 3
Evaluation harness outline	Complete	This document Section 4
Pilot plan (4-week)	Complete	This document Section 6
Priority recommendation	Complete	This document Section 7
Task generators	Not started	Week 1 build target
Scoring harness implementation	Not started	Week 1 build target
Adversarial baseline implementations	Not started	Week 2 build target
Human pilot data	Not started	Week 3 build target
Final evaluation set	Not started	Week 4 build target
Submission narrative	Not started	Week 4 build target

Related Notes

- AGI-Benchmark-Blueprint
- GGIB-Part1-Critical-Analysis
- GGIB Part 2 - Metacognition Task Family Design
- GGIB Part 3 - Formal Scoring Framework
- GGIB Part 4 - Human Baseline Design
- GGIB Part 5 - Contamination Resistance and Gaming Defenses
- Orchestrator_Epistemic_Capture_v2
- Agent_State_Framework_v3
- Metacognition-Benchmark-Critique
- Metacognition-Benchmark-Synthesis-Report

Tags

#AGI #benchmark #metacognition #kaggle-deliverables #evaluation-design #GGIB #Kaggle #competition #submission #build-plan